COMP.SE.110 GROUP PROJECT: DESIGN DOCUMENT

Tourism planner web application

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Introduction

Our group project consists of web application that combines weather data with data from events happening in near future. The purpose of the app is to both give user option to filter the events based on their weather preferences but also give idea of how they should prepare for the event in clothing and otherwise. The original idea for this application was generated by Chat GPT by first giving it the project instructions file and following prompt "Please generate few ideas for the project" of which "Tourism Planner Based on Weather and Events" was selected by the group members.

Description

When the user first arrives on the website, they are asked to either login or register an account (login feature is not shown on the prototype yet). After doing so they arrive at search page which consists of location selector, category selector, weather type selector and calendar selector. At first only location selector is enabled. After choosing one, category selector is enabled next and so on until user has successfully chosen option for each. User then starts the search by clicking a button after events that match the criteria appear marked in a calendar. User can click the circled dates on a calendar to display events as a list for that selected day. In case no event matches the criteria, information about this is given instead. User may open any of the event cards, which are shown as a list below the calendar, to display information about the event and options to view weather information for the location and the time the event is taking place. Alternatively, user may return to search page through return button.

Architecture of the application

Client-Server Architecture or Single-Page Application? – **Preferably Single Page Application**

<u>Backend:</u> Java

Currently offers /events endpoint for frontend requests.

Parameters: location, category, weather, startTime, endTime

Currently the weather parameter is not yet used for filtering the returned events.

Next some sort of /eventById endpoint will be made for getting more info about specific events.

Also weather data is yet to be linked to events. Need to figure out good way to filter events by weather forecast.

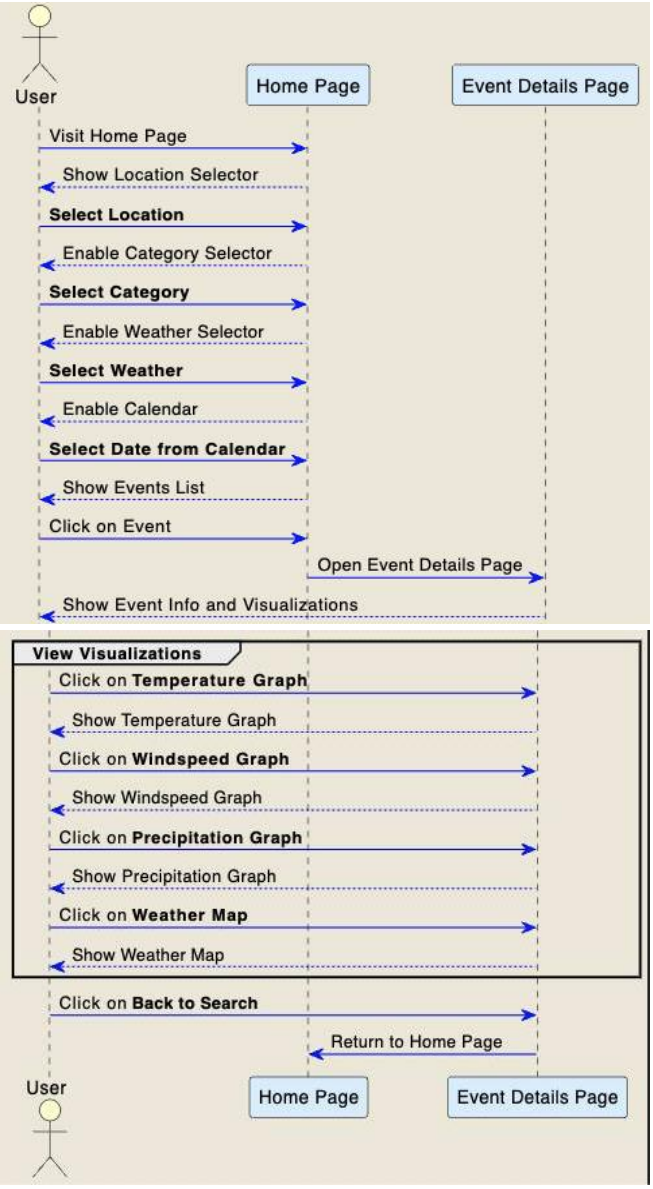
<u>UI:</u>

Language: JavaScript/TypeScript

Framework: React

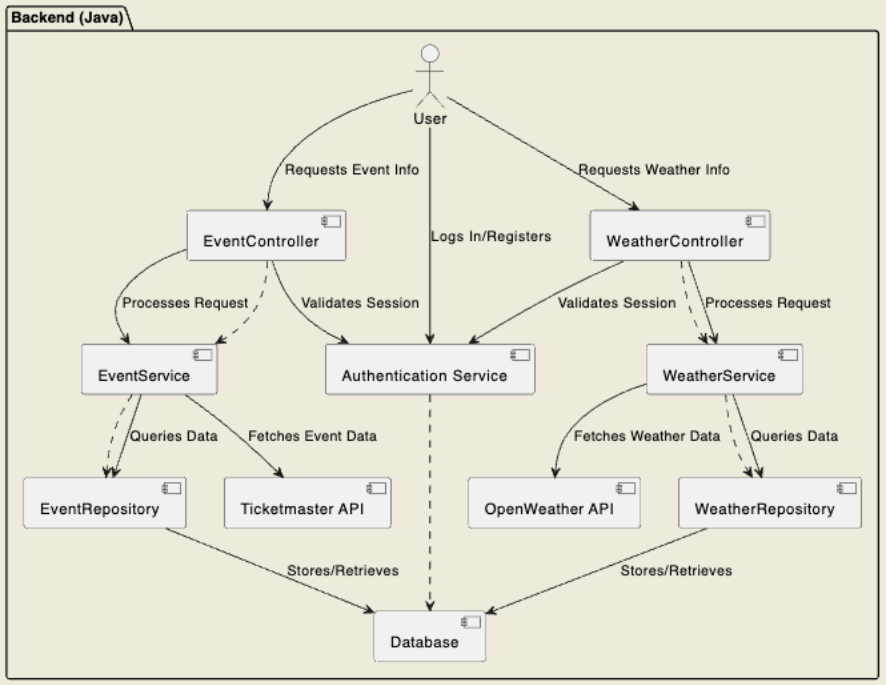
Libraries: Material UI, Leaflet, Apex Chart

Sequence Diagram:

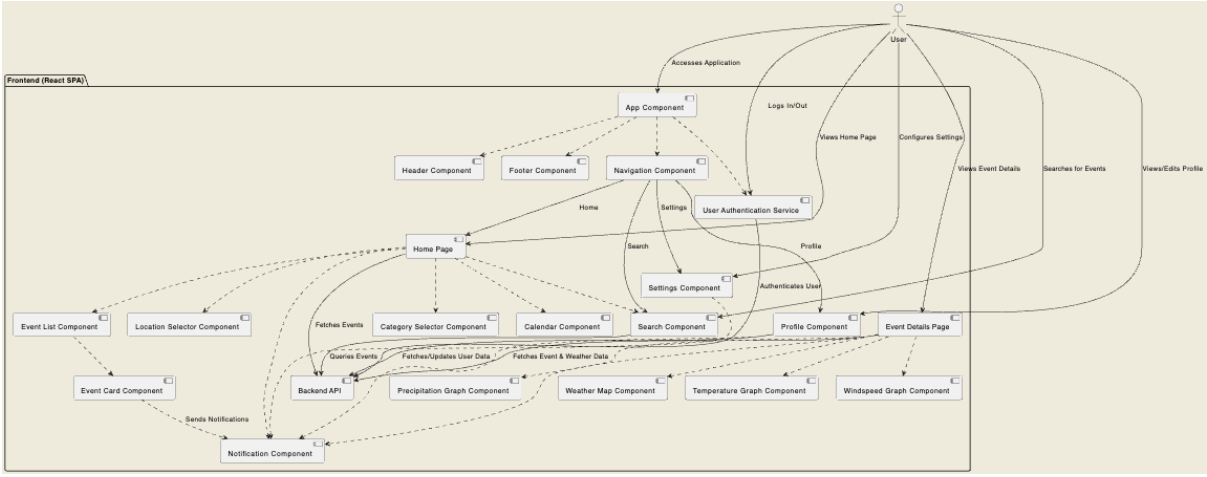


Component Diagram:

Backend:



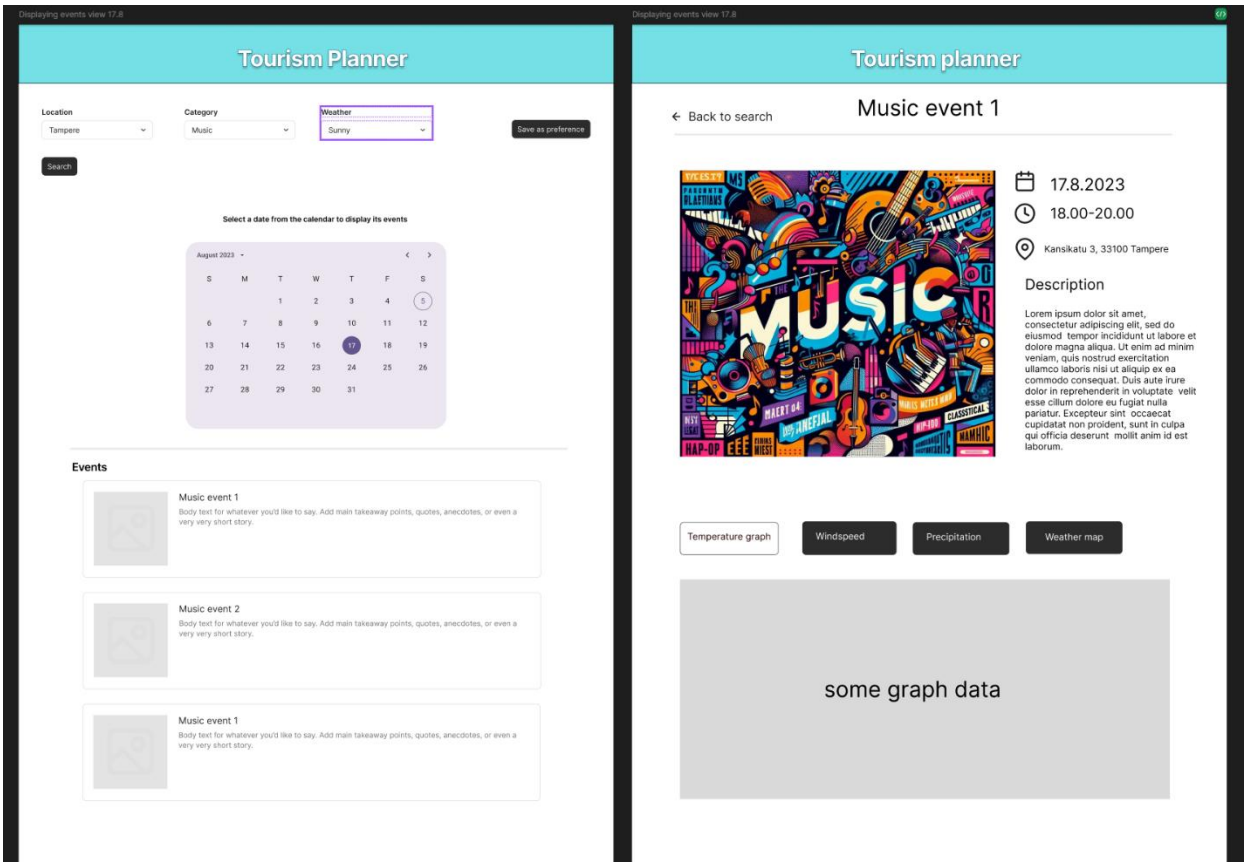
Frontend:



UI prototype:

https://www.figma.com/proto/3eTLZ5ACfWrHWrDT6vSlZb/Project-prototype-(final-version)?node-id=0-1&t=bv5hWxfP8ik0GI5I-1

Quick view of the UI prototype in case link does not work:



APIs to use

Tourism planner application needs to get weather data and data about different events, concerts, festivals and so on from APIs. We explored different open APIs and chose the ones which worked the best for this application. The selected APIs are described below.

For retrieving events data: Ticketmaster Open API

https://developer.ticketmaster.com/products-and-docs/apis/discovery-api/v2/

- Requires API key to run API calls. You can get this for free (config file or via email)
- It's possible to search for events, venues and attractions (artists, sports, packages, plays and so on) using the location or date. You can also retrieve more information of the certain event or attraction using its id.
- You can get the image of the event by using the event id.
- Classification (category) search is possible for events and attractions.
- API has rate limit of 5000 API calls per day and rate limitation of 5 requests per second.
- Return tons of information about events, need to think what all is needed

For getting weather data: OpenWeather API

https://openweathermap.org/current#name

- *Free API key data for students*
- 3,000 API calls/minute and 100,000,000 API calls/month
- *Get weather forecast for city of event*
- *Returns temperature, wind, rain, humidity, visibility etc.*
- Offers many different forecasts depending on time scale